PROJECT DESIGN PHASE-II

DATA FLOW DIAGRAM & USER STORIES

Project Title: Script-Controlled ACL – Restrict Record Access Based on Field Value

Team ID: _____

Project Name: Script-Controlled ACL – Restrict Record Access Based on Field Value

Maximum Marks: 4 Marks

1. DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) represents how information moves through the Script-Controlled ACL system. It shows the user request, record and field-value data, ACL script processing, authorization decision, and final access response.

SOURCE / ACTOR	DATA FLOW	DESTINATION / PROCESS
User	Access Request	ACL Authorization System
ACL Authorization System	Read Record + Field Value	Record Database
Record Database	Record Data + Field Value	ACL Authorization System
ACL Authorization System	User/Role + Field Value	Script-Controlled ACL
Script-Controlled ACL	Allow / Deny Decision	ACL Authorization System
ACL Authorization System	Access Granted / Access Denied	User
ACL Authorization System	Authorization Result	Audit / Log Storage

User → Access Request → Read Field Value → Script-Controlled ACL → Allow / Deny → Access Response

2. USER STORIES

The following user stories define the main functional requirements for the Script-Controlled ACL project.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
End User	Record Access	USN-1	As a user, I can request access to a protected record.	The system identifies the requested record and starts the access check	High	Sprint-1
End User	Record Access	USN-2	As a user, I can access a record when its field value satisfies the ACL rule.	The ACL script validates the field value and grants access when the rule is satisfied.	High	Sprint-1
End User	Record Access	USN-3	As a user, I cannot access a record when its field value does not satisfy the ACL rule.	The system denies access and does not expose protected record data.	High	Sprint-1
End User	Permission Response	USN-4	As a user, I can receive a clear access-denied response when I am not authorized.	A suitable permission message is displayed without revealing sensitive information	Medium	Sprint-1
Administrator	ACL Configuration	USN-5	As an administrator, I can define the field value used by the ACL condition.	The configured field is read correctly during authorization	High	Sprint-1
Administrator	ACL Script	USN-6	As an administrator, I can maintain the script that compares the field value with the permitted value.	The script executes correctly for allowed and denied cases.	High	Sprint-2
Administrator	Validation & Testing	USN-7	As an administrator, I can test allowed, denied, empty, and unexpected field-value cases.	The system produces the expected access result for each test case.	High	Sprint-2
Administrator	Audit	USN-8	As an administrator, I can review authorization results for troubleshooting.	Relevant authorization results are recorded in the audit/log area.	Medium	Sprint-2

3. EXPECTED SYSTEM BEHAVIOUR

- The system reads the configured field value from the requested record.
- The script-controlled ACL evaluates the field value against the authorization rule.
- Access is granted only when the defined condition is satisfied.
- Access is denied when the condition is not satisfied.
- The system avoids exposing protected record data to unauthorized users.
- Authorization results can be recorded for review and troubleshooting.